

5. Static

1. Write a class with 2 static variables, 2 Instance variables, 2 static methods, 2 instance methods and a main method.

Answer:

```
public class StaticProgram {

    // Static variables
    static int staticVar1 = 100;
    static int staticVar2 = 200;

    // Instance variables
    int instanceVar1 = 10;
    int instanceVar2 = 20;

    // Static method 1
    public static void staticMethod1() {
        System.out.println("This is Static Method 1");
    }

    // Static method 2
    public static void staticMethod2() {
        System.out.println("This is Static Method 2");
    }

    // Instance method 1
    public void instanceMethod1() {
        System.out.println("This is Instance Method 1");
    }

    // Instance method 2
    public void instanceMethod2() {
        System.out.println("This is Instance Method 2");
    }

    // Main method
    public static void main(String[] args) {

        // Create object
        StaticProgram obj = new StaticProgram();

        // Call methods
        staticMethod1();
        staticMethod2();

        obj.instanceMethod1();
        obj.instanceMethod2();
    }
}
```

2. Print instance variables in static methods

Answer:

```
public class PrintInstanceInStatic {
```

```

// Instance variables
int num1 = 10;
int num2 = 20;

// Static method
public static void display() {

    // Create object to access instance variables
    PrintInstanceInStatic obj = new PrintInstanceInStatic();

    // Print instance variables
    System.out.println("Instance Variable 1: " + obj.num1);
    System.out.println("Instance Variable 2: " + obj.num2);
}

// Main method
public static void main(String[] args) {

    // Call static method
    display();
}
}

```

Note: We cannot directly access instance variables inside a static method. We must create an **object**.

3. Print static variables in Instance methods

Answer:

```

public class PrintStaticInInstance {

    // Static variables
    static int number1 = 100;
    static int number2 = 200;

    // Instance method
    public void display() {

        // Print static variables
        System.out.println("Static Variable 1: " + number1);
        System.out.println("Static Variable 2: " + number2);
    }

    // Main method
    public static void main(String[] args) {

        // Create object
        PrintStaticInInstance obj = new PrintStaticInInstance();

        // Call instance method
        obj.display();
    }
}

```

Note: Instance methods can directly access **static variables**.

4. Call instance methods in static methods

Answer:

```
public class CallInstanceInStatic {

    // Instance method 1
    public void method1() {
        System.out.println("This is Instance Method 1");
    }

    // Instance method 2
    public void method2() {
        System.out.println("This is Instance Method 2");
    }

    // Static method
    public static void display() {

        // Create object
        CallInstanceInStatic obj = new CallInstanceInStatic();

        // Call instance methods
        obj.method1();
        obj.method2();
    }

    // Main method
    public static void main(String[] args) {

        // Call static method
        display();
    }
}
```

Note: We cannot directly call an **instance method** inside a **static method**. We must create an **object**.

5. Call static methods in instance methods

Answer:

```
public class CallStaticInInstance {

    // Static method 1
    public static void method1() {
        System.out.println("This is Static Method 1");
    }

    // Static method 2
    public static void method2() {
        System.out.println("This is Static Method 2");
    }

    // Instance method
    public void display() {

        // Call static methods
    }
}
```

```

        method1();
        method2();
    }

    // Main method
    public static void main(String[] args) {

        // Create object
        CallStaticInInstance obj = new CallStaticInInstance();

        // Call instance method
        obj.display();
    }
}

```

6. Print all the static, instance variables in main method

Answer:

```

public class PrintVariables {

    // Static variables
    static int staticVar1 = 100;
    static int staticVar2 = 200;

    // Instance variables
    int instanceVar1 = 10;
    int instanceVar2 = 20;

    // Main method
    public static void main(String[] args) {

        // Create object
        PrintVariables obj = new PrintVariables();

        // Print static variables
        System.out.println("Static Variable 1: " + staticVar1);
        System.out.println("Static Variable 2: " + staticVar2);

        // Print instance variables
        System.out.println("Instance Variable 1: " + obj.instanceVar1);
        System.out.println("Instance Variable 2: " + obj.instanceVar2);
    }
}

```

7. Call static methods and instance methods in main method

Answer:

```

public class CallMethods {

    // Static method 1
    public static void staticMethod1() {
        System.out.println("This is Static Method 1");
    }

    // Static method 2

```

```
public static void staticMethod2() {
    System.out.println("This is Static Method 2");
}

// Instance method 1
public void instanceMethod1() {
    System.out.println("This is Instance Method 1");
}

// Instance method 2
public void instanceMethod2() {
    System.out.println("This is Instance Method 2");
}

// Main method
public static void main(String[] args) {

    // Create object
    CallMethods obj = new CallMethods();

    // Call static methods
    staticMethod1();
    staticMethod2();

    // Call instance methods
    obj.instanceMethod1();
    obj.instanceMethod2();
}
}
```